

EARLY-STAGE DIABETES PREDICTION USING AUTOENCODER FEATURE SELECTION AND K- MEANS CLUSTERING

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

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Abstract

Diabetes prediction is treated as a binary classification problem using a synthetic Diabetes Health Indicators Dataset. The exact CSV used in this study contains 100,000 patient records and 31 columns. The workflow includes exploratory data analysis, leakage-aware preprocessing, Random Forest and XGBoost baseline classification, Autoencoder-based feature selection at 85%, 90%, and 91.05% cumulative importance thresholds, K-Means clustering on the 90% selected feature set, and hyperparameter tuning with three-fold cross-validation. A target-leakage investigation found that `diabetes_stage` had a direct relationship with `diagnosed_diabetes`, so it was excluded from predictive modelling. After leakage control, the baseline Random Forest achieved 91.98% accuracy and 0.9424 ROC-AUC. The proposed tuned pipeline, using 90% Autoencoder-selected features plus K-Means cluster information, achieved 91.02% accuracy, 91.92% F1-score, and 0.9416 ROC-AUC. The experiments show that the additional feature-selection and clustering stages preserved performance close to the baseline but did not substantially improve it on this synthetic dataset. The work demonstrates a reproducible machine-learning workflow with explicit leakage prevention and controlled experimentation.

Keywords: Diabetes Prediction, Random Forest, XGBoost, Autoencoder, K-Means Clustering, Feature Selection

1. Introduction

1.1 Purpose of the Project

Diabetes is a major health condition that requires timely identification and continuous management. Early identification of diabetes-related risk can support screening and follow-up by using information such as demographic characteristics, lifestyle habits, physical measurements, and clinical indicators. Machine learning provides a way to analyze multiple variables simultaneously and identify patterns that can be used for binary classification.

The purpose of this project is to develop an academic machine learning system for predicting diabetes from patient-related features. The project also investigates whether feature selection and patient clustering can improve or maintain predictive performance. The main stakeholders are students, researchers, and educators interested in developing reproducible machine learning workflows for healthcare-related prediction problems. Since the dataset used in this study is synthetic, the system is intended for experimentation and learning rather than direct clinical use.

1.2 Problem Statement

The problem addressed in this project is the binary classification of patients into two classes: diagnosed diabetes and no diagnosed diabetes. The input consists of demographic, lifestyle, health, and clinical features available in the dataset, while the target variable is `diagnosed_diabetes`. A value of 0 represents No Diabetes and a value of 1 represents Diabetes. Random Forest and XGBoost are used as supervised classification models, while K-Means is used as an unsupervised clustering technique to identify groups of similar patients. The expected output of the system is a predicted diabetes class together with a probability score for an unseen patient.

1.3 Existing Methods and Disadvantages

Existing diabetes prediction approaches commonly use supervised algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Gradient Boosting, and XGBoost. Ensemble

methods can capture nonlinear relationships among health-related variables, while feature-selection methods can reduce redundant inputs. Clustering approaches such as K-Means have also been explored as a way to group similar patient records before supervised prediction. Several challenges remain. Redundant or target-derived variables can increase model complexity or cause information leakage. Performance is also sensitive to preprocessing, class distribution, feature-selection strategy, and hyperparameters. In addition, high scores on synthetic or limited datasets may not translate directly to real-world populations. For these reasons, this project emphasizes leakage checking, controlled feature-selection thresholds, clustering analysis, and multiple evaluation metrics rather than accuracy alone.

1.4 Proposed System

The proposed system follows a complete pipeline: dataset inspection, EDA, preprocessing, leakage analysis, baseline RF/XGBoost, Autoencoder feature selection, K-Means clustering, hyperparameter tuning, and final evaluation. The target-derived `diabetes_stage` feature is excluded from modelling. The Autoencoder is used without the target variable to rank the 47 processed features, and three cumulative-importance thresholds are tested. The 90% set is then used for K-Means; K is selected using Elbow and Silhouette analysis. Finally, Random Forest and XGBoost are tuned using randomized search with three-fold cross-validation and evaluated on the held-out test set.

1.5 Research Objectives

1. To prepare and validate a synthetic diabetes dataset for binary classification.
2. To perform systematic data preprocessing and exploratory data analysis while preventing target leakage.
3. To develop Random Forest and XGBoost baseline models for diabetes prediction.
4. To apply Autoencoder-based feature selection using 85%, 90%, and 91.05% cumulative feature-importance thresholds.
5. To apply K-Means clustering and evaluate whether cluster information improves diabetes prediction.
6. To perform hyperparameter tuning and compare the final models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

2. Literature Survey (10 Selected Studies)

Table 1. Faculty-provided reference papers used in this project

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	The Enhancing Diabetes Prediction Accuracy Using Random Forest and XGBoost with PSO and GA-Based Feature Selection	Jawza, Mazdadi, Farmadi, Saragih, Kartini, & Abdullayev (2025), Journal of Electronics, Electromedical Engineering and Medical Informatics, 7(2), 295–306. DOI: 10.35882/jeeemi.v7i2.626	Reduce irrelevant or less useful variables while improving diabetes classification.	Test whether PSO/GA feature selection improves RF and XGBoost.	Random Forest, XGBoost, PSO, GA, AUC and confusion matrix.	Feature-selection variants changed AUC and showed that model performance depends on the selected attributes.	Evaluate larger, more diverse datasets and broader validation.
2	An Ensemble Approach to Predict Early-Stage Diabetes Risk Using Machine Learning: An Empirical Study	Laila, Mahboob, Khan, Khan, & Whangbo (2022), Sensors, 22(14), 5247. DOI: 10.3390/s22145247	Early-stage diabetes prediction requires comparison of multiple ensemble methods on a small clinical dataset.	Compare ensemble classifiers and identify useful attributes for early-stage prediction.	AdaBoost, Bagging, Random Forest, Chi-Square feature selection, multiple metrics.	Random Forest was among the strongest ensemble methods reported in the study.	Validate methods on larger and more diverse datasets.
3	An Unsupervised Cluster-Based Feature Grouping Model for Early Diabetes Detection	Hassan, Mollick, & Yasmin (2022), Healthcare Analytics, 2, 100112. DOI: 10.1016/j.health.2022.100112	Limited labeled data and irregularities make diabetes prediction difficult.	Use clustering and feature grouping before supervised prediction.	K-Means, Elbow, Silhouette, XGBoost feature ranking, RF/SVM/KNN/MLP.	Cluster-based modelling was reported to outperform modelling on the complete dataset in that study.	Test clustering on larger datasets and different populations.
4	A Deep Learning Framework with Hybrid Stacked Sparse Autoencoder for Type 2 Diabetes Prediction	Abdussamad, Daud, Sakkalingam, Zubair, Khan, & Mahmood (2025), Scientific Reports, 15, 36678. DOI: 10.1038/s41598-025-20534-4	Sparse/high-dimensional data makes feature selection and representation difficult.	Develop a sparse autoencoder framework for diabetes prediction.	Hybrid stacked sparse autoencoder, L1/L2 regularization, BCE loss, ML/DL comparisons.	Shows the potential of sparse autoencoder representations for diabetes prediction and feature selection.	Validate across more independent datasets and populations.
5	An Adaptive Ensemble Feature Selection Technique for Model-Agnostic Diabetes Prediction	Natarajan, Baskaran, & Kamalanathan (2025), Scientific Reports, 15, 6907. DOI: 10.1038/s41598-025-91282-8	Single feature selectors may be sensitive to redundancy and model choice.	Develop a model-agnostic ensemble feature-selection method.	Adaptive ensemble feature selection combining multiple selection methods and classifiers.	Combining selection methods was proposed to improve robustness over individual selectors.	Extend to low-label and semi-supervised settings.
6	Analyzing Classification and Feature Selection Strategies for Diabetes Prediction Across Diverse Diabetes Datasets	Kaliappan et al. (2024), Frontiers in Artificial Intelligence, 7, 1421751. DOI: 10.3389/frai.2024.1421751	Different diabetes datasets can favor different feature-selection and classifier combinations.	Compare classification and feature-selection strategies across diverse datasets.	RF, XGBoost, LR, GB, SVM; filter/wrapper selection; LIME/SHAP.	Shows that feature selection and explainability can materially affect diabetes-prediction workflows.	Study robustness and transferability across additional datasets.
7	Robust Predictive Framework for Diabetes Classification Using Optimized Machine Learning on Imbalanced Datasets	Abousaber, Abdallah, & El-Ghaish (2025), Frontiers in Artificial Intelligence, 7, 1499530. DOI: 10.3389/frai.2024.1499530	Class imbalance can bias learning and reduce generalization.	Develop an optimized framework for diabetes classification on imbalanced datasets.	Feature engineering, imbalance handling, optimized ML classifiers.	Highlights the importance of preprocessing and imbalance handling in reliable diabetes classification.	Validate robustness across additional real-world cohorts.
8	Identifying Diagnostic Indicators for Type 2 Diabetes Mellitus from Physical Examination Using Interpretable Machine Learning Approach	Lv et al. (2024), Frontiers in Endocrinology, 15, 1376220. DOI: 10.3389/fendo.2024.1376220	Manual exploration of large physical-examination records is difficult.	Develop interpretable screening models and identify important indicators.	Weighted diversity density models, indicator ranking, subgroup analysis.	Reported interpretable indicator rankings and a highest AUC of 0.9185.	Expand external validation and subgroup robustness.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
9	Diabetes Prediction Model Based on GA-XGBoost and Stacking Ensemble Algorithm	Li, Peng, & Peng (2024), PLoS ONE, 19(9), e0311222. DOI: 10.1371/journal.pone.0311222	Model performance can be affected by feature and parameter choices.	Optimize XGBoost and combine models through stacking.	GA-XGBoost, stacking ensemble, performance comparison.	Shows the potential benefit of optimization and ensemble combination for diabetes prediction.	Test independent datasets and improve interpretability.
10	A Deep Neural Network Prediction Method for Diabetes Based on Kendall's Correlation Coefficient and Attention Mechanism	Qi, Lu, Shi, Qi, & Ren (2024), PLoS ONE, 19(7), e0306090. DOI: 10.1371/journal.pone.0306090	Nonlinear relationships and feature relevance are difficult to capture with simpler models.	Combine correlation-based feature selection with an attention-based DNN.	Kendall correlation, polynomial-regression imputation, attention DNN, SHAP.	Reported strong test performance on two diabetes datasets and used SHAP for interpretation.	Validate on broader cohorts and assess generalization.

3. Methodology

3.1 Proposed System Flow Diagram

Dataset → Inspection → EDA → Preprocessing & Leakage Check → Baseline RF/XGBoost → Autoencoder Feature Selection → 85/90/91.05% Thresholds → K-Means → Hyperparameter Tuning → Final Evaluation → Unseen Prediction

Proposed Machine Learning Workflow



3.2 Dataset Description

The study uses the Diabetes Health Indicators Dataset from Kaggle, created and published by Mohan Krishna Thalla. The exact CSV used in this project contains 100,000 records and 31 columns. The dataset includes demographic, lifestyle, family-history, physical, cardiovascular, and laboratory indicators. The target is diagnosed_diabetes, where 0 represents No Diabetes and 1 represents Diabetes. The exact local file contains 30 columns apart from the target. Of these, diabetes_stage was excluded from predictive modelling after the leakage investigation, leaving 29 raw input features. The target distribution is 40,002 records in class 0 (40.002%) and 59,998 in class 1 (59.998%). The data is synthetic and is therefore used for academic experimentation rather than clinical diagnosis.

3.3 Experimental Setup

The project was implemented in Python using Jupyter notebooks executed through Visual Studio Code in a local virtual environment. Python 3.12.x was used. Major libraries were pandas and NumPy for data handling; Matplotlib and Seaborn for visualization; scikit-learn for preprocessing, StandardScaler, OneHotEncoder,

Random Forest, MLPRegressor-based Autoencoder, K-Means, evaluation and hyperparameter tuning; and XGBoost for gradient-boosting classification. Random state 42 was used where applicable for reproducibility.

3.4 Data Preprocessing

The dataset was inspected for data types, missing values, and duplicate records. All columns contained 100,000 non-null values and duplicate analysis returned zero duplicate rows. The target diagnosed_diabetes was separated from the inputs. The diabetes_stage feature was investigated and excluded from modelling because its categories had a direct relationship with the diagnosis target, creating a target-leakage risk. Six categorical variables were one-hot encoded and 23 numerical variables were standardized with StandardScaler. The data was split 80:20 using stratification. All fitted preprocessing transformations were learned from training data only and then applied to the test set.

3.5 Exploratory Data Analysis

EDA was performed before modelling. Descriptive statistics were calculated for numerical variables, the target distribution was visualized, numerical distributions were examined with histograms, and important clinical variables were compared across the two target classes using box plots. Categorical frequency plots were created for demographic and lifestyle variables. A numerical correlation matrix was used to identify relationships and possible redundancy. The target consisted of 59.998% diabetes cases and 40.002% no-diabetes cases. The EDA results supported systematic preprocessing and feature analysis before classification.

Categorical variables were also examined using frequency plots to identify the distribution of patient groups. In addition, a correlation matrix was generated for numerical variables to identify relationships among features and possible redundancy. The EDA results showed that the target classes were moderately imbalanced, with 59.998% of observations belonging to the diabetes class and 40.002% belonging to the non-diabetes class. The EDA also showed variation among clinical and lifestyle variables, supporting the need for systematic preprocessing and feature analysis before classification.

3.6 Feature Engineering and Feature Selection

No additional domain-derived variables were created. Categorical variables were converted by one-hot encoding and numerical variables were standardized. The leakage-prone diabetes_stage feature was removed. After preprocessing, 29 raw inputs became 47 processed features. The Autoencoder was trained without the target and used the absolute input-to-first-hidden-layer weights to derive a normalized importance score for each processed feature. Features were ranked and cumulative importance was calculated. The three requested thresholds retained 32 features at 85%, 36 features at 90%, and 37 features at 91.05%. The 90% set was carried forward because it maintained performance close to the baseline while using fewer processed inputs than the 91.05% set.

Cumulative threshold	Selected processed features	Actual cumulative importance
85%	32	85.1224%
90%	36	90.0169%
91.05%	37	91.1119%

Table 2. Autoencoder cumulative feature-importance thresholds

3.7 Model Building and Training

Random Forest and XGBoost were used as the main supervised classifiers. The baseline Random Forest used 200 trees and random_state=42. The baseline XGBoost used 200 estimators, learning_rate=0.1, max_depth=6, and logloss evaluation. For the feature-selection experiment, the Autoencoder used an MLPRegressor

architecture of 47-32-12-32-47, with the input reconstructed as the learning target. K-Means was then applied to the 90% selected features, and its cluster label was added as an additional predictor. The same models were retrained with the selected features and cluster information.

3.8 Hyperparameter Tuning

RandomizedSearchCV with three-fold cross-validation and ROC-AUC scoring was used for hyperparameter tuning. Eight random parameter combinations were evaluated for each classifier. The best Random Forest configuration was `n_estimators=266`, `max_depth=8`, `max_features=None`, `min_samples_leaf=3`, `min_samples_split=9`, `class_weight=None`, with cross-validation ROC-AUC 0.9437. The best XGBoost configuration was `n_estimators=234`, `learning_rate≈0.1174`, `max_depth=3`, `min_child_weight=3`, `subsample≈0.7512`, `colsample_bytree≈0.9356`, `gamma≈0.0998`, with cross-validation ROC-AUC 0.9420.

3.9 Model Evaluation

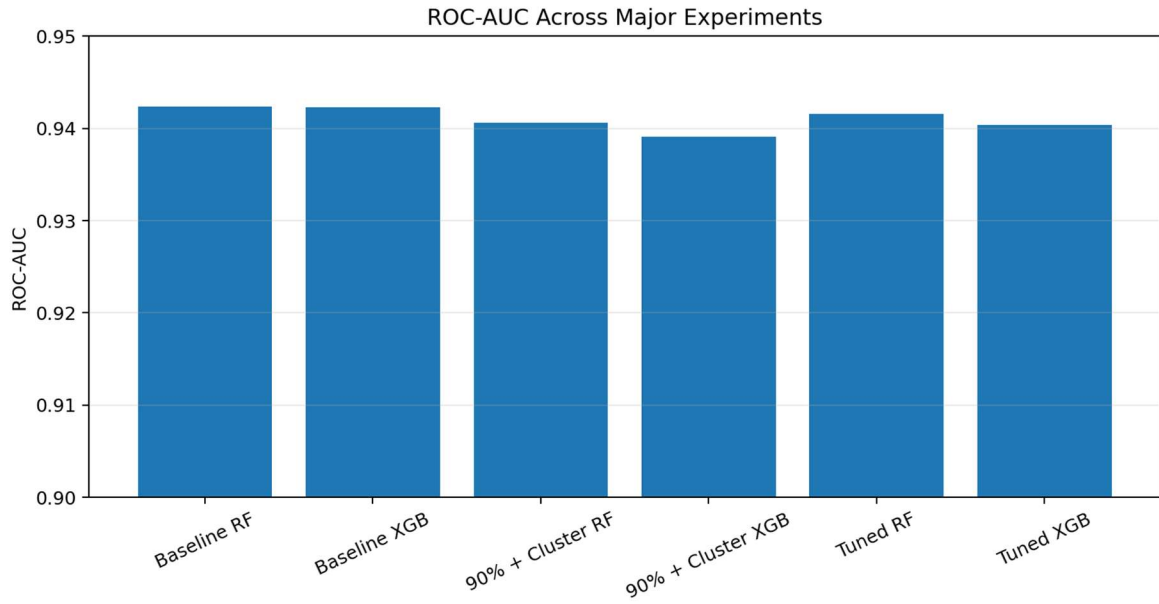
Multiple classification metrics were used: Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix, ROC curve, and classification report. Accuracy measures overall correctness; Precision measures the correctness of positive predictions; Recall measures how many actual diabetes cases are detected; F1-score balances Precision and Recall; and ROC-AUC measures discrimination between the two classes across thresholds. Final evaluation was performed on the held-out test set. Neither the target nor the test data was used during Autoencoder learning or K-Means fitting.

Confusion matrices were used to show true positives, true negatives, false positives, and false negatives. ROC curves were also generated for the classification models. Classification reports were used to summarize precision, recall, and F1-score for both classes. The final evaluation was performed on the held-out test set after preprocessing and model tuning, while the target variable was not used during Autoencoder feature learning or K-Means cluster formation.

4. Results and Discussion

4.1 Preprocessing Results

The initial dataset contained 100,000 records and 31 columns. All columns were non-null and duplicate analysis returned zero duplicates. After separating the target and excluding `diabetes_stage`, 29 raw input features remained: 23 numerical and 6 categorical. `OneHotEncoder` expanded the categorical variables and `StandardScaler` standardized numerical variables, producing 47 processed features. Stratified splitting produced 80,000 training records and 20,000 testing records. The preprocessing transformer was fitted on training data only and applied to the test set.

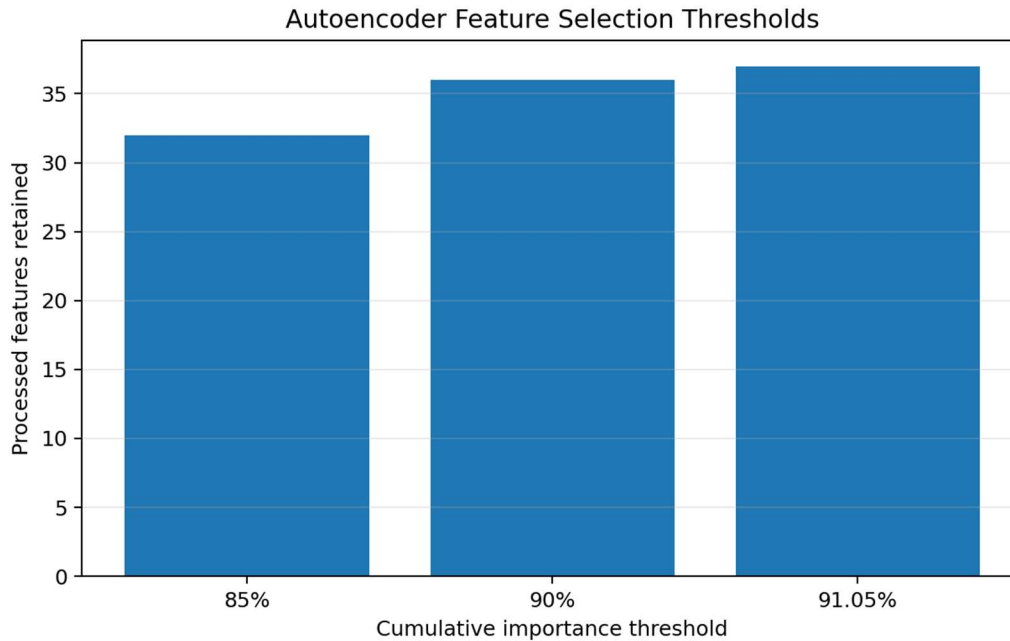


4.2 Model Performance

Table 3. Performance across the main experiments

Model / Experiment	Accuracy	Precision	Recall	F1-score	ROC-AUC
Baseline Random Forest	91.98%	99.94%	86.68%	92.84%	0.9424
Baseline XGBoost	91.93%	99.88%	86.66%	92.80%	0.9423
Autoencoder 85% + RF	61.56%	64.84%	78.50%	71.02%	0.6375
Autoencoder 85% + XGBoost	62.66%	65.20%	81.02%	72.26%	0.6481
Autoencoder 90% + RF	90.92%	99.67%	85.15%	91.84%	0.9403
Autoencoder 90% + XGBoost	90.98%	99.66%	85.27%	91.90%	0.9385
Autoencoder 91.05% + RF	90.98%	—	—	91.89%	0.9396
Autoencoder 91.05% + XGBoost	90.98%	—	—	91.89%	0.9396
90% + K-Means + RF	90.92%	99.61%	85.19%	91.84%	0.9406
90% + K-Means + XGBoost	91.00%	99.65%	85.31%	91.92%	0.9391
Tuned 90% + K-Means RF	91.02%	99.92%	85.10%	91.92%	0.9416
Tuned 90% + K-Means XGBoost	91.02%	99.83%	85.18%	91.93%	0.9404

After leakage control, both baseline classifiers achieved approximately 92% accuracy. Random Forest achieved 91.98% accuracy and ROC-AUC 0.9424, while XGBoost achieved 91.93% accuracy and ROC-AUC 0.9423. The 85% Autoencoder threshold reduced performance substantially, indicating that aggressive reduction removed information useful to the classifiers. The 90% and 91.05% thresholds retained performance much better. Adding the K-Means cluster label caused only small changes. Hyperparameter tuning raised the proposed Random Forest ROC-AUC from 0.9406 to 0.9416 and the proposed XGBoost ROC-AUC from 0.9391 to 0.9404. Across all experiments, the highest test ROC-AUC remained the leakage-free baseline Random Forest at 0.9424.

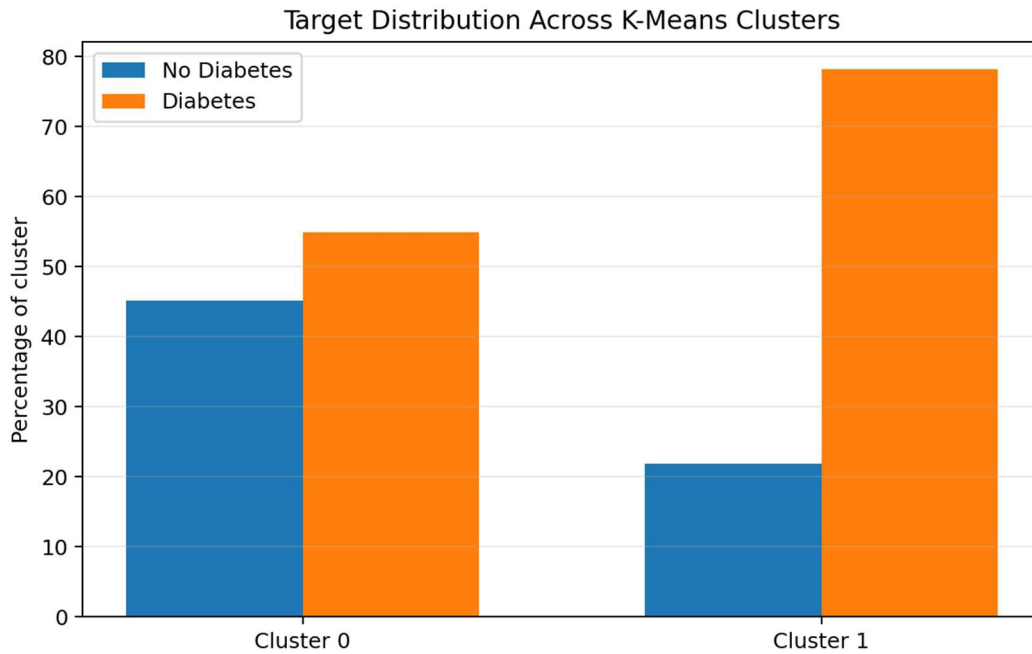


4.3 Visual Results and discussion

The visual analysis includes the Autoencoder feature-importance and cumulative-importance plots, K-Means Elbow and Silhouette plots, confusion matrices, and ROC curves. The Autoencoder required 32, 36, and 37 processed features to reach 85%, 90%, and 91.05% cumulative importance respectively. The 90% threshold therefore reduced the processed feature space by 11 features compared with using all 47 processed features.

For clustering, K values from 2 to 6 were evaluated using inertia and silhouette score. K=2 had the highest silhouette score of 0.1377. The value is modest, so the clusters were interpreted as exploratory groupings rather than strongly separated patient populations. The two training clusters contained 62,446 and 17,554 records (78.06% and 21.94%). Their diabetes proportions were 54.90% and 78.15%, showing different target distributions after clustering.

The corrected confusion matrices contain substantially more errors than the earlier leakage-affected models, highlighting the importance of excluding `diabetes_stage`. The tuned Random Forest in the proposed pipeline achieved 91.02% accuracy, 99.92% precision, 85.10% recall, 91.92% F1-score, and 0.9416 ROC-AUC. The tuned XGBoost achieved 91.02% accuracy, 99.83% precision, 85.18% recall, 91.93% F1-score, and 0.9404 ROC-AUC. These results are strong on the synthetic dataset but do not constitute clinical validation.



Final unseen-data demonstration: One held-out test record was passed through the complete preprocessing, 90% Autoencoder feature-selection, K-Means assignment, and tuned Random Forest pipeline. The actual class was 0 (No Diabetes), the predicted class was 0 (No Diabetes), and the predicted diabetes probability was 18.51%.

5. Conclusion

This project developed a complete machine-learning workflow for binary diabetes prediction using a 100,000-record synthetic dataset. After leakage control, the Random Forest baseline achieved 91.98% accuracy and 0.9424 ROC-AUC. Autoencoder feature selection was tested at 85%, 90%, and 91.05% cumulative importance; the 90% threshold retained near-baseline performance while reducing the processed feature set. K-Means selected $K=2$ using the highest silhouette score among $K=2-6$, although the modest silhouette value indicates exploratory rather than strongly separated clusters. The tuned cluster-enhanced Random Forest achieved 91.02% accuracy and 0.9416 ROC-AUC. Overall, the additional feature-selection and clustering stages did not substantially outperform the leakage-free baseline. The results demonstrate a reproducible experimental workflow, while validation on independent real-world datasets would be required before any clinical interpretation.

6. Future Work

1. Validate the proposed pipeline on independent real-world diabetes datasets.
2. Evaluate deeper, sparse, or denoising autoencoder architectures for feature selection and representation learning.
3. Investigate explainable AI techniques such as SHAP and permutation importance to improve model interpretation.
4. Explore additional clustering methods such as hierarchical clustering, DBSCAN, or Gaussian mixture models and compare cluster stability.
5. Perform broader hyperparameter optimization and repeated cross-validation to study model robustness.

6. Investigate probability calibration, decision-threshold optimization, and cost-sensitive learning, particularly for reducing missed diabetes cases.

7. References

- [1] Jawza, D. N., Mazdadi, M. I., Farmadi, A., Saragih, T. H., Kartini, D., & Abdullayev, V. (2025). Enhancing diabetes prediction accuracy using Random Forest and XGBoost with PSO and GA-based feature selection. *Journal of Electronics, Electromedical Engineering and Medical Informatics*, 7(2), 295–306. <https://doi.org/10.35882/jeeemi.v7i2.626>
- [2] Laila, U. E., Mahboob, K., Khan, A. W., Khan, F., & Whangbo, T. (2022). An ensemble approach to predict early-stage diabetes risk using machine learning: An empirical study. *Sensors*, 22(14), 5247. <https://doi.org/10.3390/s22145247>
- [3] Hassan, M. M., Mollick, S., & Yasmin, F. (2022). An unsupervised cluster-based feature grouping model for early diabetes detection. *Healthcare Analytics*, 2, 100112. <https://doi.org/10.1016/j.health.2022.100112>
- [4] Abdussamad, Daud, H., Sokkalingam, R., Zubair, M., Khan, I. K., & Mahmood, Z. (2025). A deep learning framework with hybrid stacked sparse autoencoder for type 2 diabetes prediction. *Scientific Reports*, 15, 36678. <https://doi.org/10.1038/s41598-025-20534-4>
- [5] Natarajan, K., Baskaran, D., & Kamalanathan, S. (2025). An adaptive ensemble feature selection technique for model-agnostic diabetes prediction. *Scientific Reports*, 15, 6907. <https://doi.org/10.1038/s41598-025-91282-8>
- [6] Kaliappan, J., Saravana Kumar, I. J., Sundaravelan, S., Anesh, T., Rithik, R. R., Singh, Y., Vera-Garcia, D. V., Himeur, Y., Mansoor, W., Atalla, S., & Srinivasan, K. (2024). Analyzing classification and feature selection strategies for diabetes prediction across diverse diabetes datasets. *Frontiers in Artificial Intelligence*, 7, 1421751. <https://doi.org/10.3389/frai.2024.1421751>
- [7] Abousaber, I. A., Abdallah, H. F., & El-Ghaish, H. (2025). Robust predictive framework for diabetes classification using optimized machine learning on imbalanced datasets. *Frontiers in Artificial Intelligence*, 7, 1499530. <https://doi.org/10.3389/frai.2024.1499530>
- [8] Lv, X., Luo, J., Huang, W., Guo, H., Bai, X., Yan, P., Jiang, Z., Zhang, Y., Jing, R., Chen, Q., & Li, M. (2024). Identifying diagnostic indicators for type 2 diabetes mellitus from physical examination using interpretable machine learning approach. *Frontiers in Endocrinology*, 15, 1376220. <https://doi.org/10.3389/fendo.2024.1376220>
- [9] Li, W., Peng, Y., & Peng, K. (2024). Diabetes prediction model based on GA-XGBoost and stacking ensemble algorithm. *PLoS ONE*, 19(9), e0311222. <https://doi.org/10.1371/journal.pone.0311222>
- [10] Qi, X., Lu, Y., Shi, Y., Qi, H., & Ren, L. (2024). A deep neural network prediction method for diabetes based on Kendall's correlation coefficient and attention mechanism. *PLoS ONE*, 19(7), e0306090. <https://doi.org/10.1371/journal.pone.0306090>
- [11] Thalla, M. K. (2025). Diabetes Health Indicators Dataset. Kaggle. <https://www.kaggle.com/datasets/mohankrishnathalla/diabetes-health-indicators-dataset>